

Limits (Part III) Homework

Given each function below, find the limits.

$$\textcircled{1} f(x) = \begin{cases} x^3 & x < -1 \\ \frac{1}{2}x + 1 & x \geq -1 \end{cases}$$

$$\text{i) } \lim_{x \rightarrow -1^+} f(x)$$

$$\text{ii) } \lim_{x \rightarrow -1^-} f(x)$$

$$\text{iii) } \lim_{x \rightarrow -1} f(x)$$

$$\textcircled{2} f(x) = \begin{cases} x & x \geq 2 \\ \ln x & x < 2 \end{cases}$$

$$\text{i) } \lim_{x \rightarrow 2^+} f(x)$$

$$\text{ii) } \lim_{x \rightarrow 2^-} f(x)$$

$$\text{iii) } \lim_{x \rightarrow 2} f(x)$$

$$\textcircled{3} f(x) = \begin{cases} \cos x & x \leq \pi \\ -\frac{\pi}{x} & x > \pi \end{cases}$$

$$\text{i) } \lim_{x \rightarrow \pi^+} f(x)$$

$$\text{ii) } \lim_{x \rightarrow \pi^-} f(x)$$

$$\text{iii) } \lim_{x \rightarrow \pi} f(x)$$

$$\textcircled{4} f(x) = \begin{cases} 3^{2x} & x > -\frac{1}{2} \\ 2^{x+\frac{3}{2}} - \frac{5}{3} & x \leq -\frac{1}{2} \end{cases}$$

$$\text{i) } \lim_{x \rightarrow -\frac{1}{2}^+} f(x)$$

$$\text{ii) } \lim_{x \rightarrow -\frac{1}{2}^-} f(x)$$

$$\text{iii) } \lim_{x \rightarrow -\frac{1}{2}} f(x)$$

Find the limits.

⑤ i) $\lim_{x \rightarrow 1^+} \frac{|x-1|}{x-1}$

ii) $\lim_{x \rightarrow 1^-} \frac{|x-1|}{x-1}$

iii) $\lim_{x \rightarrow 1} \frac{|x-1|}{x-1}$

⑥ i) $\lim_{x \rightarrow -2^+} \frac{x+2}{|x+2|}$

ii) $\lim_{x \rightarrow -2^-} \frac{x+2}{|x+2|}$

iii) $\lim_{x \rightarrow -2} \frac{x+2}{|x+2|}$

⑦ $\lim_{x \rightarrow 0} \left[\frac{\cos x - 1}{x} - \frac{8 \sin x}{x} \right]$

⑧ $\lim_{x \rightarrow 4} 6$

⑨ $\lim_{x \rightarrow 0} \sin\left(\frac{1-\cos x}{x}\right)$

⑩ $\lim_{x \rightarrow 0} \left[\frac{(1-\cos x) \sin x}{x^2} \right]$

"e" is a constant.

⑪ $\lim_{x \rightarrow -7} (-e)$

⑫ $\lim_{x \rightarrow 0} \frac{\frac{\sin x}{x}}{x^2+11}$

⑬ $\lim_{x \rightarrow 0} \left[\frac{2 \cos x - 1}{5x} \right]$

" π " and "e" are constants.

⑭ $\lim_{x \rightarrow -\sqrt{6}} (4e - 2\pi + \sqrt{7})$

⑮ $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^3$

$$(16) \lim_{x \rightarrow 0} \left[\csc(x+1) + \frac{\cos x - 1}{x} \right]$$

$$(17) \lim_{x \rightarrow 0} \left[\frac{\sin^2 x}{x^2} \right]$$

$$(18) \lim_{x \rightarrow 0} \left(\frac{\frac{1 - \cos x}{x}}{x^2 + 3x - 9} \right)$$

$$(19) \lim_{x \rightarrow 0} \left[\log_3(x+8) + \frac{\sin x}{x} \right]$$

If $\lim_{x \rightarrow a} f(x) = 8$, $\lim_{x \rightarrow a} g(x) = -1$ and $\lim_{x \rightarrow a} h(x) = \frac{1}{10}$, find the value of each expression. Write the number of the limit law used on each step.

$$(20) \lim_{x \rightarrow a} [100f(x)]$$

$$(21) \lim_{x \rightarrow a} \left[\frac{g(x)}{f(x)} \right]$$

$$(22) \lim_{x \rightarrow a} [h(x)f(x)]$$

$$(23) \lim_{x \rightarrow a} \left[f(x) - \frac{h(x)}{4} \right]$$

$$\lim_{x \rightarrow a} f(x) = 8, \lim_{x \rightarrow a} g(x) = -1, \lim_{x \rightarrow a} h(x) = \frac{1}{10}$$

$$(24) \lim_{x \rightarrow a} [\log_2(f(x))]$$

$$(25) \lim_{x \rightarrow a} x^2$$

$$(26) \lim_{x \rightarrow a} [-9 + h(x)]$$

$$\lim_{x \rightarrow a} f(x) = 8, \lim_{x \rightarrow a} g(x) = -1, \lim_{x \rightarrow a} h(x) = \frac{1}{10}$$

$$(27) \lim_{x \rightarrow a} \left[\frac{f(x) + h(x)}{4} \right]$$

$$(28) \lim_{x \rightarrow a} [x^2 f(x)]$$

Find the limits algebraically.

$$(29) \lim_{t \rightarrow 9} \left[\frac{-2}{t-9} + \frac{3t+9}{t^2-81} \right]$$

$$(30) \lim_{y \rightarrow -\frac{1}{2}} \left[\frac{6y+3}{4y^2+2y} - \frac{1}{y} \right]$$

$$\textcircled{31} \lim_{m \rightarrow \frac{1}{2}} \left[\frac{1}{2m-1} + \frac{1}{8m^2-10m+3} - \frac{1}{4m-3} \right]$$

Homework Answers

① i) $\frac{1}{2}$

iii) $\frac{1}{3}$

ii) -1

⑤ i) 1

iii) $\boxed{\text{DNE}}$

ii) -1

② i) 2

iii) $\boxed{\text{DNE}}$

ii) $\ln 2$

⑥ i) 1

iii) $\boxed{\text{DNE}}$

ii) -1

iii) $\boxed{\text{DNE}}$

③ i) -1

ii) -1

⑦ $2 \& 5$ -8

iii) -1

⑧ 6 6

④ i) $\frac{1}{3}$

⑨ 7 0

ii) $\frac{1}{3}$

⑩ 3 0

⑪ 6 $-e$

⑫ 3 $\frac{4}{5}$

⑫ $4 \& 8$ $\frac{1}{11}$

⑬ $2 \& 5$ $7\frac{39}{40}$ or $\frac{319}{40}$ or 7.975

⑬ 5 0

⑭ 7 3

⑭ 6 $4e^{-2\pi} + \sqrt{7}$

⑮ 8 a^2

⑮ 7 1

⑯ $1 \& 6$ $-8\frac{9}{10}$

⑯ $1 \& 8$ $\csc(1)$

⑰ $4 \& 1 \& 6$ $\frac{81}{40}$ or $2\frac{1}{40}$ or 2.025

⑰ 3 1

or $5 \& 1$

⑱ $4 \& 8$ 0

⑱ $3 \& 8$ $8a^2$

⑲ $1 \& 8$ 5

⑳ $\frac{1}{18}$

㉑ 5 800

㉑ -1

㉒ 4 $-\frac{1}{8}$

㉒ -1